MA388 Sabermetrics: Lesson 24

Streakiness Statistics

```r
library(tidyverse)
library(knitr)
library(baseballr)
```

Lesson Objectives

1. Calculate a clumpiness/streakiness coefficient (sum of squared gap lengths) for any player.
2. Use simulation to generate the null distribution of the streakiness coefficient under randomness.
3. Compare a player's observed streakiness to the simulated null distribution.
4. Identify the longest hitless streaks among MLB players and interpret them.

Review - DiMaggio's Streak

Last class we talked about the incredible hitting streak of Joe DiMaggio. In 1941, Joe DiMaggio of the New York Yankees hit safely (got at least one hit) in 56 straight games. The streak began on May 15th, 1941 when DiMaggio went 1-for-4 against the Chicago White Sox and ended on July 17th when he went 0-for-3 with a walk against the Cleveland Indians.

Individual At-Bat Streaks

We looked at DiMaggio's streak of games with hits, but now we turn our attention to streaks in individual at-bats. In 2011, Miguel Cabrera led the league with a batting average of 0.344. Let's take a look at his streaks for the season.

```r
# Load the data.

retro2011 <- retrosheet_data(years_to_acquire = '2011') |>
```

```
  pluck('2011') |>
  pluck('events')
```

Filter the data to include only Miguel Cabrera's plate appearances.

```
library(Lahman)
cabrera_id <- People |>
  filter(nameFirst == "Miguel", nameLast == "Cabrera") |>
  select(retroID) |> pull()

cabrera <- retro2011 |>
  subset(bat_id == cabrera_id & ab_fl==TRUE)
```

To look at streaks we need to first identify which at-bats resulted in a hit and which resulted in an out.

```
cabrera <- cabrera |>
  mutate(H = ifelse(h_fl > 0, 1, 0),
         DATE = str_sub(game_id, 4, 12),
         AB = 1,
         AB_Num = row_number()) |>
  arrange(DATE)
```

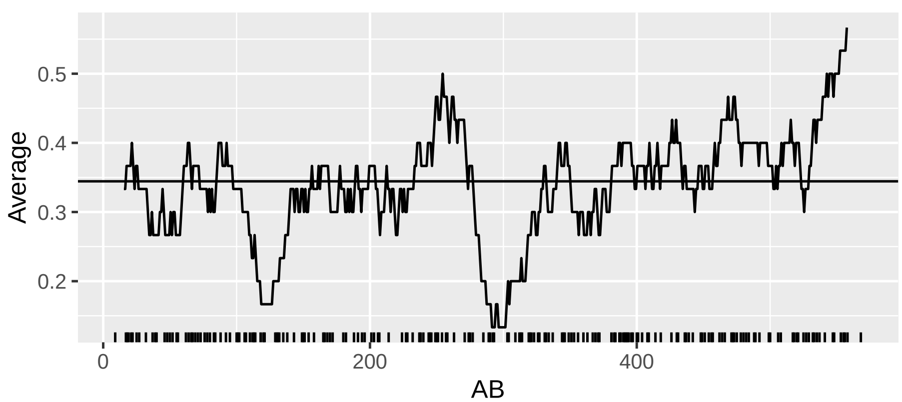
We can then use the streaks function we defined last class to calculate the length of hit streaks and out streaks.

```
streaks <- function(y){
  x <- rle(y)
  class(x) <- "list"
  return(as_tibble(x))
}

# First, let's look at hit streaks.
cab_hits <- cabrera |>
  pull(H) |>
  streaks() |>
  filter(values == 1)

cab_hits |>
  pull(lengths)
```

```
  [1] 1 1 2 1 3 1 4 1 1 2 1 1 2 1 1 3 1 1 1 1 1 1 3 1 1 1 1 1 1 1 1 1 1 2 2 1 1
 [38] 1 1 1 1 1 2 1 2 1 1 2 1 2 1 1 2 1 2 1 3 1 1 3 3 3 1 2 1 1 1 1 1 2 1 1 2 2
 [75] 1 1 1 1 1 2 2 2 1 3 2 1 2 2 1 1 1 1 2 1 1 1 1 2 2 1 1 1 1 3 1 1 1 3 1 1 1
[112] 1 1 1 1 1 1 3 2 1 1 1 1 2 1 1 1 1 1 1 2 2 5 2 2 1 2
```

What can you learn from looking at Cabrera's hitting streaks over the season?

```r
library(zoo)
moving_average <- function(df, width){
  N <- nrow(df)
  df |>
    transmute(Game = rollmean(1:N, k = width, fill = NA),
              Average = rollsum(H, width, fill = NA) / rollsum(AB, width, fill = NA))
}

# Let's assume ~3 ABs/game, so 30 ABs will give us a ~10-game window.
moving_average(cabrera, 30) |>
  ggplot(aes(Game, Average)) +
  geom_line() + xlab("AB") +
  geom_hline(yintercept = mean(cabrera$H)) +
  geom_rug(data = cabrera |> filter(H==1),
           aes(AB_Num, .3 * H), sides = "b")
```



After looking at the plot, does it change your conclusions from before?

Now let's look at Cabrera's hitting slumps.

```
# Find Cabrera's hitting slumps.
cab_outs <- cabrera |>
  pull(H) |>
  streaks() |>
  filter(values == 0)

cab_outs |>
  pull(lengths)
```

```
  [1]  1  3  2  6  1  7  2  2 10  5  1  2  1  1  5  2  3  6  2  1  1  2  1  1  1
 [26] 14  5  2  3  1  4  4  3  1  3  3  2  2  1  4  1  2  6  1  5  1  2  4  1  1
 [51]  8  2  3  4  2  1  3  1  5  3  2  2  2  4  7  9  5  3  7  1  7  3  1  4  1
 [76]  1  4  1  2  4  9  4  2  4  1  3  4  1  3  3  2  2  2  1  3  1  2  3  1  2
[101]  5  3  4  1  2  1  2  2  1  1  2  3  2  3  1  1  2  3  1  8  1  1  1  1  1
[126]  1  1  6  2  1  2  1  1  2  2  3
```

What can you learn from looking at Cabrera's hitting slumps over the season?

How does this compare to the rest of the league? Since Cabrera had 572 at bats over the season, let's compare him to other players who had over 500 at-bats.

```
# Filter the data to get all players with over 500 at-bats in the season.
retro_500 = retro2011 |>
  group_by(bat_id) |>
  summarise(AB = sum(ab_fl)) |>
  filter(AB >= 500) |>
  pull(bat_id)

# Create a function to calculate the longest slump for each player.
longest_ofer <- function(batter){
  retro2011 |>
    filter(bat_id == batter, ab_fl == TRUE) |>
    mutate(H = ifelse(h_fl > 0, 1, 0),
           DATE = substr(game_id, 4, 12)) |>
    arrange(DATE) |>
    pull(H) |>
```

```r
    streaks() |>
    filter(values == 0) |>
    summarize(MAX_STREAK = max(lengths))
}

# Find each player's longest slump.
retro_500 |>
  map_df(longest_ofer) |>
  mutate(BAT_ID = retro_500) |>
  left_join(People |> select(nameFirst, nameLast, retroID),
            by = c("BAT_ID" = "retroID")) |>
  arrange(-MAX_STREAK) |>
  mutate(Rank = row_number()) |>
  knitr::kable()
```

MAX_STREAK	BAT_ID	nameFirst	nameLast	Rank
35	ibanr001	Raul	Ibanez	1
30	aybae001	Erick	Aybar	2
27	mcgec001	Casey	McGehee	3
24	sancg001	Gaby	Sanchez	4
23	betay001	Yuniesky	Betancourt	5
23	bourp001	Peter	Bourjos	6
23	hardj003	J. J.	Hardy	7
23	howar001	Ryan	Howard	8
23	lee-c001	Carlos	Lee	9
23	ortid001	David	Ortiz	10
23	pennc001	Cliff	Pennington	11
23	santc002	Carlos	Santana	12
22	escoa003	Alcides	Escobar	13
22	reynm001	Mark	Reynolds	14
22	riosa002	Alex	Rios	15
22	walkn001	Neil	Walker	16
21	kinsi001	Ian	Kinsler	17
20	cabra002	Asdrubal	Cabrera	18
20	cuddm001	Michael	Cuddyer	19
20	huffa001	Aubrey	Huff	20
20	ramia001	Aramis	Ramirez	21
20	trumm001	Mark	Trumbo	22
20	uptoj001	Justin	Upton	23
20	victs001	Shane	Victorino	24
20	younc004	Chris	Young	25
19	desmi001	Ian	Desmond	26
19	gonza002	Alex	Gonzalez	27
19	granc001	Curtis	Granderson	28

MAX_STREAK	BAT_ID	nameFirst	nameLast	Rank
19	huntt001	Torii	Hunter	29
19	jacka001	Austin	Jackson	30
19	jonea003	Adam	Jones	31
19	matsh001	Hideki	Matsui	32
19	maybc001	Cameron	Maybin	33
19	rollj001	Jimmy	Rollins	34
19	swisn001	Nick	Swisher	35
19	valed001	Danny	Valencia	36
19	younm003	Michael	Young	37
18	bonie001	Emilio	Bonifacio	38
18	butlb003	Billy	Butler	39
18	kendh001	Howie	Kendrick	40
18	markn001	Nick	Markakis	41
18	stubd001	Drew	Stubbs	42
18	teixm001	Mark	Teixeira	43
18	uptob001	B. J.	Upton	44
18	wietm001	Matt	Wieters	45
18	zobrb001	Ben	Zobrist	46
17	barnd001	Darwin	Barney	47
17	brucj001	Jay	Bruce	48
17	fielp001	Prince	Fielder	49
16	abreb001	Bobby	Abreu	50
16	bartj001	Jason	Bartlett	51
16	escoy001	Yunel	Escobar	52
16	espid001	Danny	Espinosa	53
16	gardb001	Brett	Gardner	54
16	hosme001	Eric	Hosmer	55
16	johnk003	Kelly	Johnson	56
16	mccua001	Andrew	McCutchen	57
16	pedrd001	Dustin	Pedroia	58
16	philb001	Brandon	Phillips	59
16	ramia003	Alexei	Ramirez	60
16	suzui001	Ichiro	Suzuki	61
16	wellv001	Vernon	Wells	62
15	andre001	Elvis	Andrus	63
15	beltc001	Carlos	Beltran	64
15	crawc002	Carl	Crawford	65
15	damoj001	Johnny	Damon	66
15	ellsj001	Jacoby	Ellsbury	67
15	freef001	Freddie	Freeman	68
15	guerv001	Vladimir	Guerrero	69
15	hilla001	Aaron	Hill	70
15	lonej001	James	Loney	71
15	pradm001	Martin	Prado	72

MAX_STREAK	BAT_ID	nameFirst	nameLast	Rank
14	cabrm001	Miguel	Cabrera	73
14	gonza003	Adrian	Gonzalez	74
14	jeted001	Derek	Jeter	75
14	peraj001	Jhonny	Peralta	76
14	stanm004	Giancarlo	Stanton	77
14	tulot001	Troy	Tulowitzki	78
14	uggld001	Dan	Uggla	79
14	vottj001	Joey	Votto	80
14	wertj001	Jayson	Werth	81
13	casts001	Starlin	Castro	82
13	infao001	Omar	Infante	83
13	kempm001	Matt	Kemp	84
13	pench001	Hunter	Pence	85
13	pujoa001	Albert	Pujols	86
12	braur002	Ryan	Braun	87
12	crisc001	Coco	Crisp	88
12	fukuk001	Kosuke	Fukudome	89
12	konep001	Paul	Konerko	90
12	kotcc001	Casey	Kotchman	91
12	morsm001	Mike	Morse	92
12	pierj002	Juan	Pierre	93
11	cabrm002	Melky	Cabrera	94
11	canor001	Robinson	Cano	95
11	franj004	Jeff	Francoeur	96
11	gorda001	Alex	Gordon	97
10	bautj002	Jose	Bautista	98
10	bourm001	Michael	Bourn	99
10	martv001	Victor	Martinez	100
9	reyej001	Jose	Reyes	101

A Streakiness Statistic

We can quantify streakiness by looking at the sum of the squares of the gaps between successive hits (called the “streakiness” or “clumpiness” coefficient).

Calculate the streakiness coefficient for Cabrera.

```
# Streakiness Coefficient
cabrera_coeff <- cabrera |>
  pull(H) |>
  streaks() |>
  filter(values == 0) |>
```

```
  summarize(C = sum(lengths ^ 2)) |>
  pull()

cabrera_coeff
```

```
[1] 1681
```

Now we can simulate what may have happened at random for a player with the same number of hits and outs as Cabrera.

```
# Create a function to simulate the streakiness coefficient.
random_mix <- function(y) {
  y |>
    sample() |>
    streaks() |>
    filter(values == 0) |>
    summarize(C = sum(lengths ^ 2)) |>
    pull()
}

# Run the simulation 10000 times.
cabrera_random <- replicate(10000, random_mix(cabrera$H))
```

Plot the simulation results.

```
ggplot(data.frame(cabrera_random), aes(cabrera_random)) +
  geom_histogram(aes(y = stat(density)), bins = 20,
                 color = "blue", fill = "white") +
  geom_vline(xintercept = cabrera_coeff, size = 2) +
  annotate(geom = "text", x = cabrera_coeff * 1.05,
           y = 0.0010, label = "OBSERVED", size = 5) +
  theme_minimal() +
  labs(x = "Clumpiness Statistic",
       y = "Density")
```

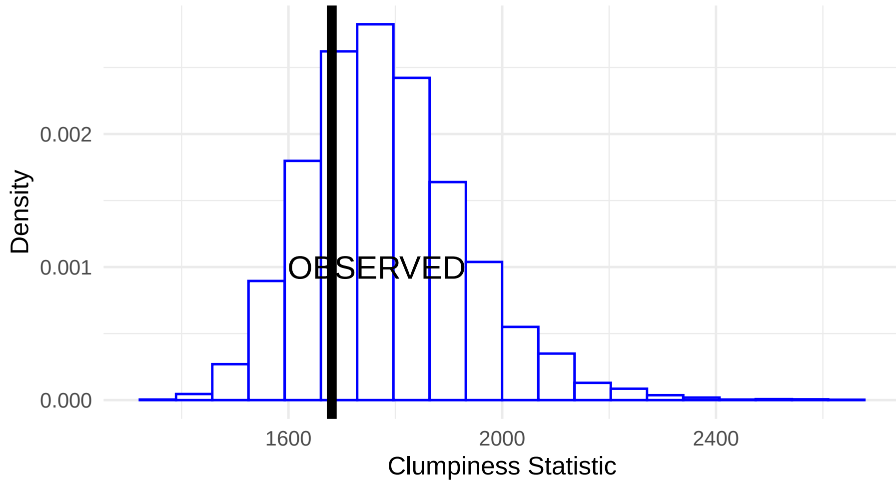


Figure 1: Histogram of ten thousand values of the clumpiness statistic assuming all arrangements of hits and outs for Cabrera are equally likely. The observed value of the clumpiness statistic for Cabrera is shown using a vertical line.

How do we interpret the plot?